

Physics-Guided Flow-Map Matching for Precipitation Nowcasting

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Abstract. Precipitation nowcasting, generating future radar fields from past observations, is critical for flood warning and disaster response. It is also a demanding benchmark for spatiotemporal generative modeling, with chaotic dynamics, heavy-tailed intensities, and rare high-intensity structures that matter most. Deterministic models minimize a pixel loss and are driven toward the conditional mean, which blurs exactly those structures, while generative models that add a stochastic residual on top of a deterministic backbone inherit the same blur. We propose Physics-Guided Flow-Map Matching (PG-FMM), a conditional flow-map model that decouples predictable advection from uncertain small-scale detail. A frozen Lagrangian advection prior transports the radar field and supplies an explicit motion forecast, and a flow-map generative head, conditioned on the past frames and the prior rollout rather than summed onto it, produces sharp stochastic detail in four sampling steps. The prior serves only as guidance, so the head replaces blurred structure instead of inheriting it. Extensive experiments on four radar benchmarks show that PG-FMM outperforms state-of-the-art methods on 18 of 24 metrics, with the largest gains at heavy-rain thresholds, where the critical success index improves by up to 58.9%. Code and pretrained checkpoints are available at <https://github.com/Neurogica/PG-FMM>.

Keywords: Flow matching · Generative models · Spatio-temporal modeling · Video prediction · Precipitation nowcasting

1 Introduction

Precipitation nowcasting generates the next one to two hours of high-resolution reflectivity fields from a short history of radar frames. It is conditional video prediction in a demanding regime, as the dynamics are chaotic, the intensity distribution is heavy-tailed, and forecast value is concentrated in rare high-intensity structures. Nowcasts underpin flood early warning, disaster response, water management, and the safe operation of transport and energy infrastructure [2, 43, 60],

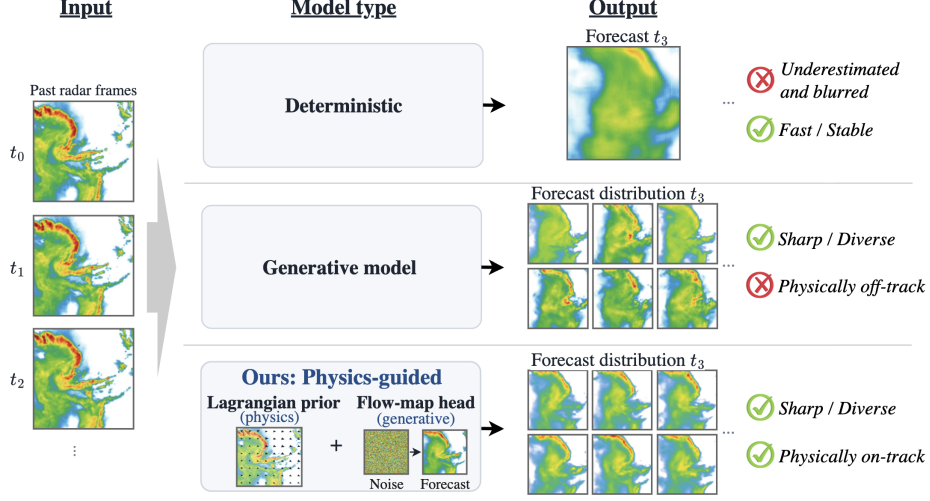


Fig. 1: Three families of precipitation nowcasters: (1) deterministic models blur high-intensity cells; (2) purely generative models can drift off the observed storm track; (3) our physics-guided design conditions a flow-map head on a frozen advection prior, keeping forecasts sharp and on track.

and their importance grows as a warming climate intensifies short-duration convective rainfall extremes and the flash flooding they cause [12]; the short lead time of a nowcast often separates an effective response from a costly one. A useful nowcast must therefore forecast where and how strongly convective cells will develop, not merely whether it will rain.

This high-intensity regime is also the hardest, because small-scale convective growth and decay are only partly predictable. Deterministic video predictors minimize a per-pixel regression loss and are therefore driven toward the conditional mean of the future, so their forecasts blur [14, 16, 46, 47, 54]. The blur erases sharp, high-intensity cores, and detection skill at the heaviest thresholds decays toward zero as the lead time grows. This failure mode is shared even by strong recent predictors such as PhyDNet [26] and AlphaPre [28], because a mean-seeking objective smooths extremes regardless of the backbone. A faithful nowcaster must therefore be generative, sampling sharp and plausible fields rather than averaging them.

Existing generative nowcasters address sharpness but leave two gaps. Many recent diffusion-based methods anchor the forecast on a deterministic backbone, refining it or adding a stochastic residual to it [18, 59]; because that backbone is trained with a pixel loss, it carries conditional-mean blur, which the anchored forecast inherits at long lead times. Purely generative models, including GAN- and flow-based nowcasters [39, 43], avoid the anchor but discard the predictable part of the dynamics, namely the advective motion that a simple physical model already captures well, and must relearn it from data, which can drift the forecast

off the observed storm track. These gaps motivate a design that retains the motion skill of a physical model while letting a generative model render the unpredictable detail, without either contaminating the other. Figure 1 contrasts deterministic, purely generative, and our physics-guided designs.

In this paper we propose Physics-Guided Flow-Map Matching (PG-FMM), a generative nowcaster that guides a flow-map generative head with a frozen physical motion model. A lightweight Lagrangian advection prior is pre-trained once and then frozen; it rolls the observed field forward through a learned velocity and source term and supplies an explicit motion forecast. A flow-map generative head then produces the future sequence conditioned on both the history and this prior rollout. A central design choice is that the prior enters as a conditioning signal rather than as an additive residual, so the head can replace the prior’s blurred structure with sharper detail while still exploiting its motion skill, and the physical rollout keeps the convection on a plausible trajectory. Here *physics-guided* means that the prior’s rollout is a discretized advection equation in which only the coefficient fields are learned (Sec. 4.2). Casting the generator as a two-time flow map rather than an instantaneous velocity field lets it produce a forecast in a few network evaluations, in line with recent flow-map and consistency models [6, 48]. This separation of predictable motion from stochastic detail follows the decoupling logic proposed in [34].

Across four standard radar benchmarks and six metrics each, PG-FMM outperforms the state-of-the-art AlphaPre [28] on 18 of the 24 metrics, and the margins are largest precisely at the heavy-rain thresholds where mean-seeking methods fail. Code and pretrained checkpoints are publicly available.⁵ The main contributions of this work are summarized as follows.

- We propose PG-FMM, which guides a generative flow-map head with a frozen Lagrangian advection prior used as conditioning rather than an additive residual, so conditional-mean blur is never inherited additively.
- We introduce, to our knowledge, the first flow-map-matching formulation for precipitation nowcasting, coupling a learned semi-Lagrangian advection prior with a two-time transport head whose few-step sampling makes the inference-time ensemble affordable.
- We show on the SEVIR, MeteoNet, Shanghai, and CIKM benchmarks that PG-FMM surpasses AlphaPre on 18 of 24 metrics, with the largest gains at the heavy-rain thresholds, and we isolate the prior’s contribution with controlled ablations, which show that its benefit stems from the physics content rather than from conditioning alone.

2 Related Work

2.1 Video Prediction and Precipitation Nowcasting

Radar nowcasting is a conditional video-prediction problem, and the two literatures have co-evolved since ConvLSTM [46], which was introduced for now-

⁵ <https://github.com/Neurogica/PG-FMM>

casting and became a standard video-prediction baseline [33, 36]. Deterministic sequence predictors developed on video transfer directly to radar, including trajectory-aware recurrence (TrajGRU [47], the PredRNN family [53–55], MotionRNN [56]), efficient convolutional designs (SimVP [14], MAU [8]), and transformer and operator backbones (Earthformer [16], Rainformer [4]); physically structured predictors such as PhyDNet [26] and the recent amplitude–phase model AlphaPre [28], the strongest deterministic predictor on the benchmarks we use, continue this line, and surveys cover both the video and the weather perspective [2, 45]. Because a squared- or absolute-error objective is minimized by the conditional mean, all of these models tend to blur, and the blur is most damaging at high reflectivity where sharp convective cells are smoothed away.

Stochastic video generation addresses the blur by sampling futures rather than averaging them, as in video-diffusion predictors such as MCVD [52] and STRPM [7]. On radar, DGMR [43] uses a conditional GAN, and NowcastNet [60] couples a deterministic evolution operator with a generative network. Diffusion and latent-diffusion models have since become dominant. PreDiff [15] and LD-Cast [27] perform latent diffusion, CasCast [18] cascades a deterministic prediction with a latent diffusion refiner, and DiffCast [59] adds a stochastic residual on top of an arbitrary deterministic backbone. We refer to the latter recipe as a residual anchor: the forecast is a deterministic prediction plus a generated residual. Because the deterministic part is trained with a pixel loss, it carries conditional-mean blur that the sum cannot remove, particularly at long lead times. Diffusion models for the broader weather-forecasting problem [35, 40, 50] share the same generative motivation at coarser scales.

2.2 Flow Matching and Flow Maps

Continuous-time generative models have matured rapidly, and recent surveys [30, 58] cover the progression from score-based diffusion [19, 21, 49] to simulation-free transport. Flow Matching [29], Rectified Flow [32], and Stochastic Interpolants [1] regress an instantaneous velocity field whose probability-flow ODE must be integrated with many steps at sampling time. Flow-Map Matching [6] instead learns the finite-time flow map (the two-time transport operator), so that a sample is produced in a few network evaluations, or even one; Consistency Models [48], Consistency Trajectory Models [22], and Shortcut Models [13] reach the same few-step regime through self-consistency, trajectory matching, and step-size conditioning, and distillation offers a complementary route [44]; data-to-data diffusion bridges [31, 61] are related transport formulations. In nowcasting, FlowCast [39] applies conditional flow matching in a latent space. We adopt a flow-map head because its few-step sampling makes a multi-member inference ensemble affordable, which is otherwise prohibitive for long diffusion chains.

2.3 Physics-Informed and Lagrangian Forecasting

Physics-informed machine learning injects physical knowledge into data-driven models through soft PDE-residual penalties, structured operators, or explicit

dynamics, and is reviewed comprehensively by Karniadakis *et al.* [20, 42]. In precipitation nowcasting, a long operational tradition extrapolates the radar field along estimated motion, as in the optical-flow libraries PySTEPS [41] and rainymotion [3]; such Lagrangian advection is accurate for short horizons but cannot represent growth or decay. Learning-based work reintroduces the missing source term while keeping the motion explicit: NowcastNet [60] learns a neural evolution operator with an additive intensity update, PhyDNet [26] disentangles physical dynamics from residual factors in a latent space, and the differentiable Lagrangian network LUPIN [38] separates advection from an advection-free growth/decay branch. Beyond nowcasting, large data-driven weather models [5, 23, 24, 37] extend physically informed architectures to global forecasting. Our prior follows the explicit warp-and-source route but keeps the physics module lightweight and, after a short pre-training stage, frozen, so that it supplies motion skill as a conditioning signal, not a constraint on the generative head.

3 Preliminaries

3.1 Problem Setting

We address radar precipitation nowcasting. A model observes a history of T_{in} past radar frames and predicts the next T_{out} frames, where each frame is a single-channel reflectivity field $R \in \mathbb{R}^{H \times W}$ on a spatial grid of size $H \times W$ normalized to $[0, 1]$. We write the observed history as $R_{1:T_{\text{in}}}$ and the prediction target as

$$x_0 = R_{T_{\text{in}}+1:T_{\text{in}}+T_{\text{out}}} \in \mathbb{R}^{T_{\text{out}} \times H \times W}, \quad (1)$$

where the T_{out} future frames are stacked along the channel axis for the convolutional networks. The horizon lengths $T_{\text{in}}, T_{\text{out}}$, the grid size $H \times W$, and the per-dataset intensity scale used to recover physical units are dataset-specific and are listed in Sec. 5.

Because convective growth and decay are only partly determined by the recent past, the future is inherently uncertain. We therefore model the conditional distribution $p(x_0 \mid R_{1:T_{\text{in}}})$ and draw an ensemble of forecasts from it rather than predicting a single field. A useful forecast combines high detection skill at the operational intensity thresholds, especially the heavy-rain thresholds, with low pixel error and sharp structure. We report the Critical Success Index (CSI) per threshold and averaged, the Heidke Skill Score (HSS), SSIM, and MSE, with formal definitions deferred to Sec. 5; the primary metrics are the mean CSI and the high-threshold CSI.

3.2 Flow Matching and Flow Maps

Continuous-time generative models transport a simple base density to the data distribution along an interpolant between a noise sample $x_1 \sim \mathcal{N}(0, I)$ and a data sample x_0 ,

$$x_\tau = (1 - \tau) x_0 + \tau x_1, \quad (2)$$

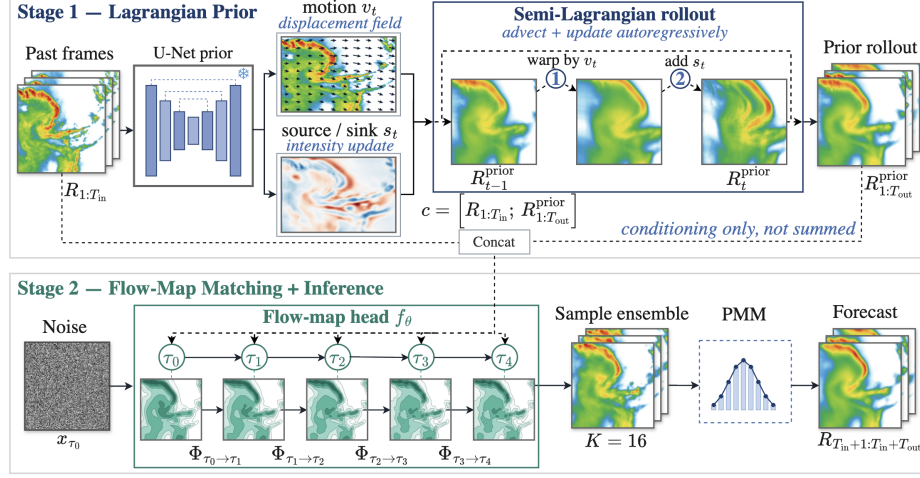


Fig. 2: Overview of PG-FMM. A frozen Lagrangian advection prior g_{ϕ} produces an explicit motion forecast, and a flow-map head f_{θ} , conditioned on the history and the prior rollout, maps noise to the future sequence in a few transport steps. Stage 1 is pre-trained and frozen, Stage 2 is trained with the flow-map objective, and PMM denotes the probability-matched mean over K samples. The prior is never added to the output.

where $\tau \in [0, 1]$ indexes the path from data ($\tau=0$) to noise ($\tau=1$). Flow matching learns an instantaneous velocity field $v_{\theta}(x_{\tau}, \tau)$ whose probability-flow ODE carries the base density to the data density, and sampling integrates this ODE, which typically requires many function evaluations.

Flow-Map Matching instead learns the two-time solution operator of the same ODE. The flow map $\Phi_{t \rightarrow r}$ takes a state at time t directly to the corresponding state at an earlier time r ,

$$\Phi_{s \rightarrow r} \circ \Phi_{t \rightarrow s} = \Phi_{t \rightarrow r}, \quad (3)$$

where $0 \leq r < s < t \leq 1$ and the semigroup identity holds because a flow map is a solution operator. Enforcing this composition consistency during training regularizes the learned operator, and because the map spans a finite time interval a sample is produced in a few jumps rather than many small steps. Our generative head (Sec. 4) instantiates a conditional flow map of this form.

4 Method

4.1 Overview

PG-FMM forecasts precipitation by separating the predictable advective motion of the radar field from the uncertain small-scale detail, so that neither part contaminates the other. The model has two modules (Fig. 2): a frozen Lagrangian

advection prior g_ϕ (Sec. 4.2) that turns the observed history into an explicit motion forecast, and a flow-map generative head f_θ (Sec. 4.3) that, conditioned on the history and the prior rollout, samples the future sequence. At inference we aggregate a small ensemble with a probability-matched mean (Sec. 4.4).

The prior is used only as conditioning and is never added to the output, unlike residual-anchor nowcasters [18, 59], whose pixel-loss backbone carries its conditional-mean blur into the anchored forecast (Sec. 2). Passing the physical forecast as a conditioning signal instead lets the generative head replace the prior’s blurred structure with sharp detail while still exploiting its motion skill. The decoupling does not depend on the particular modules, as any per-step motion model and any conditional generative head could be substituted. Ours pairs a learned warp-and-source prior with a flow-map head whose few-step sampling keeps the ensemble affordable.

4.2 Lagrangian Advection Prior

Precipitation evolves partly through predictable transport and partly through stochastic growth and decay, and a lightweight physical model can already account for the first part. The prior captures this predictable motion explicitly, so the generative head models only the residual uncertainty rather than relearning bulk advection from scratch. The prior is implemented as a U-Net g_ϕ that ingests the stacked history $R_{1:T_{\text{in}}}$ and emits, for each future step t , a dense velocity field $v_t \in \mathbb{R}^{2 \times H \times W}$ and a scalar source/sink field $s_t \in \mathbb{R}^{H \times W}$. These two fields are the only learned quantities of Stage 1; the rollout operator below contains no parameters. Both outputs are bounded for stability as

$$v_t = d_{\max} \tanh(\tilde{v}_t), \quad s_t = c_s \tanh(\tilde{s}_t), \quad (4)$$

where $\tilde{v}_t, \tilde{s}_t$ are the raw network outputs, $d_{\max}$ caps the per-step displacement, and c_s caps the intensity update. Starting from the last observed frame, the forecast is rolled out autoregressively as

$$R_t^{\text{prior}} = \text{clip}_{[0,1]} \left(\text{warp} (R_{t-1}^{\text{prior}}, v_t) + s_t \right), \quad t = 1, \dots, T_{\text{out}}, \quad R_0^{\text{prior}} = R_{T_{\text{in}}}, \quad (5)$$

where $\text{warp}(\cdot, v)$ is backward bilinear sampling and $\text{clip}_{[0,1]}$ keeps intensities in the normalized range. The update is a semi-Lagrangian discretization of the advection equation with a source term, $\partial_t R + v \cdot \nabla R = s$. The warp evaluates each output pixel at its upstream location, displaced backward along v_t , which is the Lagrangian view of tracking precipitation parcels along the flow, so existing intensity is transported across the grid rather than re-synthesized; the additive source s_t models the growth and decay that pure advection cannot represent. Learning is thus confined to the coefficient fields v_t and s_t . The rollout operator itself has no parameters, so every forecast of the prior satisfies the discretized advection equation by construction.

Stage 1 is supervised regression rather than generative modeling. The prior is trained alone, before the generative head, to forecast the observed future frames

under the composite objective

$$\mathcal{L}_{\text{prior}} = \ell_1^w + \lambda_{\text{mse}} \text{MSE} + \lambda_{\text{csi}} \mathcal{L}_{\text{soft-CSI}} + \lambda_v \|\nabla v\|_1 + \lambda_s \|s\|_1, \quad (6)$$

where $\ell_1^w = \mathbb{E}[(1+2R)|\hat{R} - R|]$ is an intensity-weighted ℓ_1 distance between the rollout $\hat{R} = R_t^{\text{prior}}$ and the observed future frame R (mean over pixels, frames, and the batch), $\mathcal{L}_{\text{soft-CSI}}$ is a differentiable sigmoid relaxation of the CSI detection metric of Sec. 5 (construction in the supplementary material), $\|\nabla v\|_1$ and $\|s\|_1$ penalize rough motion fields and overactive sources so that advection explains as much of the change as possible, and λ_{mse} , λ_{csi} , λ_v , and λ_s are scalar weights (values in the supplementary material). After this stage the prior is frozen and receives no gradient during generative training or at inference.

4.3 Flow-Map Matching Generative Head

A deterministic prior cannot supply sharp, calibrated detail, so the second module is generative. It models the conditional distribution of the future frames given the history and the prior rollout, and it is realized as a flow map rather than an instantaneous velocity field, so that sampling needs only a few network evaluations and the inference ensemble (Sec. 4.4) remains affordable. We treat the future sequence x_0 as the data endpoint of a linear stochastic interpolant [1] between Gaussian noise $x_1 \sim \mathcal{N}(0, I)$ and data, as in Eq. (2). The conditioning c concatenates, along the channel axis, the observed history $R_{1:T_{\text{in}}}$ and the frozen prior rollout $R_{1:T_{\text{out}}}^{\text{prior}}$ from Eq. (5); it is appended to the network input at every evaluation and, again, never added to the output.

Instead of regressing the instantaneous velocity of the interpolant, the network learns its flow map: $f_\theta(x_t, c; t, r)$ predicts the state x_r directly from the state x_t for $0 \leq r < t \leq 1$ [6]; here t and r are interpolant times, not frame indices. The two times enter through a timestep embedding and as two constant channel planes. The backbone is a U-Net with self-attention at its coarsest scale; its width and depth are reported in the supplementary material. The primary loss regresses the operator to the true interpolant endpoint as

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, x_1, (t, r)} \|f_\theta(x_t, c; t, r) - x_r\|_2^2, \quad (7)$$

where the times satisfy a minimum gap $t - r \geq \delta_{\text{min}}$ and, with probability p_0 , we force $r=0$ so the network also learns the direct noise-to-data map used at the first sampling step. A single large jump regresses toward a conditional average; multi-step sampling (Sec. 4.4) and the consistency term below counteract this. So that the learned map behaves as a transport operator, we add a composition-consistency regularizer that enforces the semigroup identity of Eq. (3), writing $\Phi_{t \rightarrow r}(\cdot) \equiv f_\theta(\cdot, c; t, r)$ with the conditioning c shared by every factor. For a random intermediate time $s \in (r, t)$, composing $t \rightarrow s$ and then $s \rightarrow r$ must

reproduce the direct map $t \rightarrow r$; violations are penalized as

$$\begin{aligned} \mathcal{L}_{\text{CC}} = & \underbrace{\frac{1}{2}\|\Phi_{t \rightarrow s}(x_t) - x_s\|_2^2 + \frac{1}{2}\|\Phi_{s \rightarrow r}(\Phi_{t \rightarrow s}(x_t)) - x_r\|_2^2}_{\text{sub-map fidelity}} \\ & + \underbrace{\|\Phi_{s \rightarrow r}(\Phi_{t \rightarrow s}(x_t)) - \Phi_{t \rightarrow r}(x_t)\|_2^2}_{\text{self-consistency}}, \end{aligned} \quad (8)$$

where x_s is the interpolant state at the intermediate time s (Eq. (2)), the expectation is over x_0, x_1 and times $r < s < t$ drawn as in Eq. (7), and the intermediate prediction is detached before the second map; the two fidelity terms coincide with Eq. (7) evaluated at (t, s) and (s, r) . The total objective is $\mathcal{L}_{\text{FM}} + \lambda_{\text{CC}} \mathcal{L}_{\text{CC}}$, with the schedule for λ_{CC} given in the supplementary material; in our setting the consistency term stabilizes training and improves fidelity.

4.4 Inference: Few-Step Sampling and Ensemble

Because f_θ is a transport operator, a forecast is produced with a small number of function evaluations (NFE). On a decreasing time grid $1=\tau_0 > \tau_1 > \dots > \tau_N=0$, we initialize $x_{\tau_0} \sim \mathcal{N}(0, I)$ and iterate as

$$x_{\tau_{i+1}} = f_\theta(x_{\tau_i}, c; \tau_i, \tau_{i+1}), \quad i = 0, \dots, N-1, \quad (9)$$

returning $\text{clip}_{[0,1]}(x_{\tau_N})$, where N is the number of sampling steps; each step is one evaluation of f_θ , so a forecast costs exactly N NFE. We use exponential-moving-average weights of the trained network. A single draw is sharp, but its small-scale detail varies with the noise seed. Few-step sampling makes it inexpensive to draw an ensemble of K independent samples from different noise seeds and combine them with a probability-matched mean (PMM) [11]. Averaging the members directly would blur again, since displaced peaks cancel; the PMM instead takes the spatial pattern of the ensemble mean, which places precipitation reliably, and remaps its values, rank by rank, onto the value distribution pooled over all K members, so the high-amplitude peaks that plain averaging smooths away are restored, which is decisive for high-threshold CSI. We report the full PMM for verification metrics and, for qualitative figures, a single sample, which is visually sharper; the operating-point values N and K are given in Sec. 5.

5 Experiments

5.1 Datasets and Protocol

We evaluate on four standard radar benchmarks spanning different regions and sensors: SEVIR (United States; vertically integrated liquid, VIL) [51], MeteoNet (France) [25], Shanghai Radar (China) [9], and CIKM [10]. Following the AlphaPre protocol [28], every frame is resized to $H \times W = 128 \times 128$ and normalized to $[0, 1]$, and we predict T_{out} future frames from $T_{\text{in}}=5$ past frames, with $T_{\text{out}}=20$

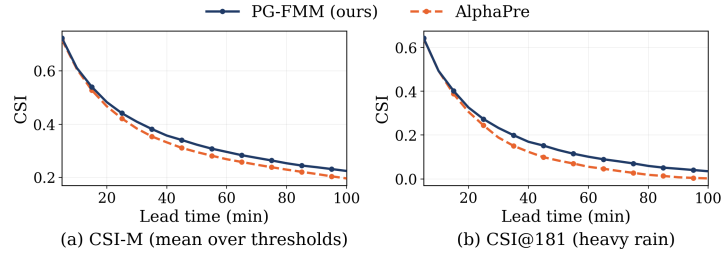


Fig. 3: Detection skill versus lead time on SEVIR: CSI averaged over thresholds (a) and at the heavy-rain threshold $VIL=181$ (b). The margin of PG-FMM over AlphaPre widens with lead time and is largest on heavy rain.

on SEVIR, MeteoNet and Shanghai and $T_{\text{out}}=10$ on CIKM. We use each benchmark’s official splits throughout: SEVIR provides 12,182 training and 8,096 test sequences; MeteoNet 6,308 / 1,310; Shanghai 1,381 / 526 (a 10% tail of the official 1,534 training sequences is held out for validation, with no test leakage); and CIKM 1,000 / 4,000. Each dataset keeps its native intensity scale, thresholds and evaluator, so all numbers are comparable.

5.2 Implementation Details

The two modules are trained in sequence with AdamW (learning rate 10^{-4} , batch size 8) and exponential-moving-average weights (decay 0.999). The Lagrangian prior is trained first and then frozen; the flow-map head is then trained for 297.6k steps for the headline SEVIR model, and every ablation variant uses a matched 150k-step budget. The prior U-Net uses base width 96 with channel multipliers (1, 2, 4, 4); the flow-map U-Net uses base width 128 with multipliers (1, 2, 3, 4) and self-attention at the two coarsest scales. At inference we use $N=4$ sampling steps and a $K=16$ -member probability-matched-mean ensemble, i.e., 64 network evaluations per forecast (1.2 forecasts/s on one GPU; a single sample runs at 7.2/s). Training and evaluation run on one NVIDIA GB202 GPU (96 GB), and checkpoints are selected by validation skill. Full hyperparameters, per-dataset training steps, and timing are listed in the supplementary material.

5.3 Baselines and Metrics

We compare against the baselines reported in the AlphaPre study under the same protocol. These include the deterministic predictors ConvGRU [47], MAU [8], SimVP [14], FourCastNet [37], Earthformer [16], PhyDNet [26], and Earthfarseer [57], and the strongest recent deterministic and generative methods, NowcastNet [60], DiffCast [59], and AlphaPre [28].

We report the standard meteorological detection and fidelity metrics. For a binarization threshold κ , counting hits (TP), misses (FN), and false alarms (FP)

Table 1: Comparison on four radar benchmarks under the AlphaPre evaluation protocol [28]; ours uses a 16-member probability-matched mean. ND/D denotes models without/with an explicit dynamics module, and CSI- x is the CSI at threshold x , the dataset’s two heavy-rain levels. In each column, the best result is in **bold** and the second best is underlined.

Model	Type	SEVIR						MeteoNet					
		CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-24 $\uparrow$	CSI-32 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	ND	0.2903	0.0879	0.0350	0.3619	0.6100	368.34	0.3401	0.2990	0.1431	0.4667	0.7833	12.85
MAU	ND	0.3076	0.1071	0.0516	0.3863	0.6505	355.48	0.3233	0.2839	0.0997	0.4452	0.7897	12.92
SimVP	ND	0.3108	0.1106	0.0517	0.3924	0.6508	383.56	0.3351	0.3002	0.1130	0.4573	0.7804	13.45
FourCastNet	ND	0.2686	0.0717	0.0339	0.3355	0.5976	410.27	0.3027	0.2533	0.1085	0.4216	0.6450	15.05
Earthformer	ND	0.2892	0.0844	0.0245	0.3665	0.6633	360.11	0.3205	0.2884	0.1237	0.4491	0.7772	14.43
PhyDNet	D	0.3017	0.1040	0.0278	0.3812	0.6532	357.63	0.3384	0.3194	0.1366	0.4673	0.7823	14.48
Earthfarseer	D	0.3004	0.0992	0.0413	0.3829	0.6327	388.91	0.3404	0.3170	0.1372	0.4726	0.7542	14.10
NowcastNet	D	0.2791	0.0770	0.0351	0.3512	0.6839	412.94	0.3427	0.3206	0.1598	0.4751	0.7879	15.64
DiffCast	D	0.3050	0.1300	<u>0.0582</u>	0.3996	0.6482	559.59	0.3512	0.3340	0.1808	0.4846	0.7887	17.93
AlphaPre	D	<u>0.3259</u>	<u>0.1332</u>	0.0545	<u>0.4110</u>	<u>0.6884</u>	<u>345.18</u>	<u>0.3824</u>	<u>0.3633</u>	<u>0.2002</u>	<u>0.5164</u>	<u>0.7968</u>	<u>12.74</u>
Ours	D	0.3614	0.1859	0.0925	0.4593	0.7291	315.67	0.4239	0.4066	0.2293	0.5583	0.8426	9.25

Model	Type	Shanghai						CIKM					
		CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CSI-M $\uparrow$	CSI-35 $\uparrow$	CSI-40 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$
ConvGRU	ND	0.3612	0.3163	0.2062	0.4899	0.7796	33.56	0.3091	0.2009	0.1259	0.4006	0.6507	37.13
MAU	ND	0.3983	0.3621	0.2417	0.5346	0.7195	30.40	0.3039	0.2054	0.1241	0.3928	0.6325	40.74
SimVP	ND	0.3850	0.3549	0.2382	0.5194	0.7795	34.40	0.3052	0.2044	0.1321	0.3955	0.6538	38.06
FourCastNet	ND	0.3571	0.3108	0.2073	0.4868	0.5598	32.10	0.2980	0.1849	0.1015	0.3801	0.4359	<u>36.14</u>
Earthformer	ND	0.3503	0.3178	0.1872	0.4844	0.7298	35.57	0.3077	0.2039	0.1369	0.4001	0.6267	36.49
PhyDNet	D	0.3654	0.3236	0.2176	0.4957	0.7751	36.41	0.3038	0.2052	0.1287	0.3931	0.6541	39.56
Earthfarseer	D	0.3926	0.3608	0.2343	0.5330	0.5405	32.68	0.3000	0.2046	0.1259	0.3911	0.6373	39.87
NowcastNet	D	0.3953	0.3608	0.2450	0.5334	0.7902	33.56	0.2991	0.1940	0.1188	0.3865	0.6713	40.96
DiffCast	D	0.4089	0.3740	0.2606	0.5476	0.7879	36.35	<u>0.3159</u>	0.2009	<u>0.1457</u>	0.4085	0.6499	42.78
AlphaPre	D	<u>0.4178</u>	0.3854	<u>0.2615</u>	0.5534	<u>0.7951</u>	<u>28.02</u>	0.3194	<u>0.2068</u>	0.1416	0.4137	<u>0.6568</u>	35.18
Ours	D	0.4181	<u>0.3819</u>	0.2636	<u>0.5520</u>	0.7962	26.05	0.3149	0.2218	0.1540	<u>0.4115</u>	0.6252	42.18

over the thresholded prediction and ground truth, the Critical Success Index and Heidke Skill Score are

$$\text{CSI}_\kappa = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}, \quad \text{HSS} = \frac{2(\text{TP} \text{TN} - \text{FP} \text{FN})}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})}, \quad (10)$$

where TN counts correct negatives. CSI-M is the mean CSI over the dataset’s thresholds, and we additionally report CSI at the two highest (heavy-rain) thresholds, the hardest regime. We also report SSIM and MSE for pixel fidelity, following the benchmark protocol; SSIM rewards smoothness, so we treat it as a reference value, and the primary metrics are CSI-M and the high-threshold CSI. As our model predicts a distribution, we additionally report the continuous ranked probability score (CRPS) [17], a strictly proper scoring rule computed from the 16-member ensemble, for our model and its generative ablation variants, which share one operating point; the Tab. 1 baselines are transcribed from the AlphaPre benchmark, which defines no CRPS protocol.

5.4 Main Results

Across the four benchmarks and six metrics each, PG-FMM outperforms AlphaPre on 18 of the 24 metrics (Tab. 1). It is best on every metric on SEVIR and MeteoNet, and the gains concentrate at the heavy-rain thresholds: on the heaviest SEVIR threshold CSI rises from 0.0582 to 0.0925, a 58.9% relative

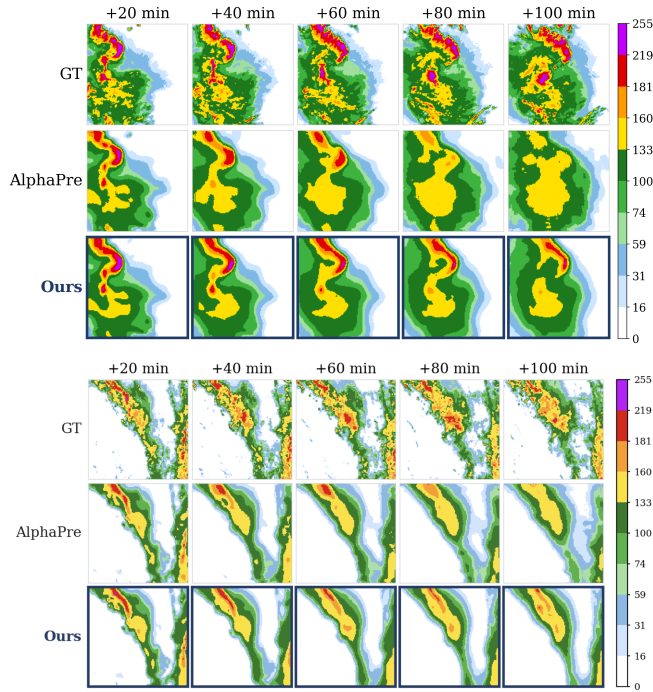


Fig. 4: Qualitative forecasts on SEVIR-VIL (rows: ground truth, AlphaPre, PG-FMM; columns: +20 to +100 minutes). Top: a convective system over Nebraska (test index 1708). Bottom: the remnants of Hurricane Barry (15 July 2019). PG-FMM preserves the heavy-rain cores and peak intensities that AlphaPre blurs.

gain. Figure 3 traces this margin over lead time, the CSI-M gap widening from +0.006 at +5 minutes to +0.028 at +100 minutes while the baseline’s heavy-rain detection decays toward zero. On Shanghai, PG-FMM is best on four of six metrics, including CSI-40, SSIM, MSE and, narrowly, mean CSI; on CIKM it is best on the two heavy-rain thresholds and second on HSS. PG-FMM also outperforms the strongest generative baseline, DiffCast, on every SEVIR metric, and its 16-member ensemble attains a CRPS of 0.0294 on SEVIR, with a spread–skill diagnostic in the supplementary material. Sampling-seed variability at the reported operating point is negligible, far below the reported margins (supplementary material).

5.5 Qualitative Results

Figure 4 shows two cases over the full 100-minute horizon (single samples, the sharper qualitative view; verification uses the probability-matched mean, Sec. 4.4). On the Nebraska convective system (top), PG-FMM preserves the sharp, high-intensity core at every lead time while AlphaPre progressively attenuates and blurs it; the frozen prior keeps the convection on its observed track

Table 2: Prior ablation on SEVIR (16-member PMM, four sampling steps), rows sorted by CSI-M; the generative variants differ only in the prior content, and Prior only additionally removes the generative head; we omit its CRPS, which degenerates to the pixel MAE for a deterministic forecast and is not comparable to ensemble CRPS. In each column, the best result is in **bold** and the second best is underlined.

Prior	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CRPS $\downarrow$
Classical advection	0.3373	0.1490	0.0745	0.4260	0.7113	327.13	0.0308
None (pure generative)	0.3525	0.1690	0.0884	0.4466	0.6968	319.60	0.0312
Prior only (no generative head)	0.3552	0.1737	0.0859	0.4493	0.7330	334.95	–
Deterministic (AlphaPre)	<u>0.3571</u>	<u>0.1782</u>	<u>0.0892</u>	<u>0.4530</u>	<u>0.7244</u>	<u>317.87</u>	<u>0.0299</u>
Lagrangian physics (ours)	0.3601	0.1807	0.0907	0.4573	0.7226	315.28	0.0298

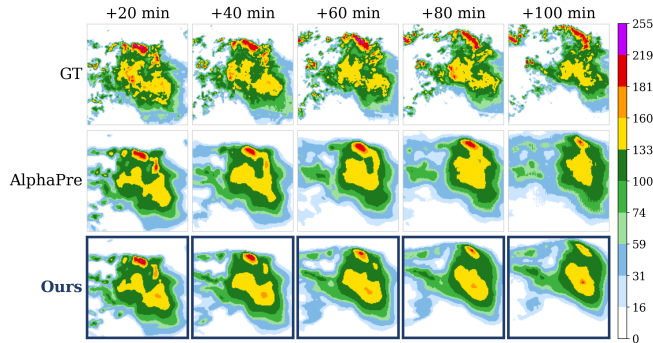


Fig. 5: A representative failure case, a convective system over north Texas (test index 103; rows and colors as in Fig. 4). With growing lead time PG-FMM drops the heavy rain band that both the ground truth and AlphaPre retain (CSI-160 0.218 vs. 0.299).

while the generative head restores intensity. The same behavior holds on the remnants of Hurricane Barry (bottom).

To test whether these examples are representative rather than favorable, we ranked *all* 2,672 heavy-rain SEVIR test samples (ground-truth max VIL ≥ 120) by the per-sample CSI-160 gain over AlphaPre at the reported operating point; AlphaPre attains the higher CSI-160 on 351 of them (13.1%). Figure 5 shows a representative failure, a convective system over north Texas where the generative head drops the heavy-rain band that both the ground truth and the deterministic baseline retain. The supplementary material provides additional cases on SEVIR and MeteoNet.

5.6 Ablation Study

Four generative variants share an identical flow-map head, identical conditioning, and a matched 150k-step training budget on SEVIR, and differ only in the content of the prior rollout (Tab. 2): no prior, classical semi-Lagrangian advection driven by optical flow, a learned deterministic nowcaster (AlphaPre) used as the prior, and our frozen Lagrangian physics prior. The physics prior is best on every

metric except SSIM, followed by the learned deterministic prior, and classical advection trails even the no-prior variant on every detection metric. The content of the prior is therefore what matters: a poor prior is worse than no prior at all, so the gains cannot be explained by conditioning alone. The one exception, SSIM, rewards smoother output, as the Prior only row below shows most clearly. The full model in Tab. 1 uses a longer schedule, hence its slightly higher numbers; per-lead-time curves in the supplementary material show that the physics prior’s margin emerges at long lead times.

The generative head is likewise necessary. The frozen prior alone (Prior only in Tab. 2) is already a strong deterministic baseline, yet it trails the full model on every skill metric; its best-in-table SSIM reflects smoothness, and the head adds the sharpness and probabilistic skill (CRPS). Together, these ablations justify the motion–generative decoupling.

The supplementary material reports further studies. Composition consistency (Eq. (8)) appears to improve fidelity and stability rather than detection skill, so we keep it enabled. An ensemble-size sweep supports the 16-member operating point, and even a single draw ($K=1$, CSI-M 0.3394) at $N=4$ network evaluations exceeds AlphaPre’s published CSI-M. A fusion ablation indicates that conditioning, unlike additive residual fusion, is robust to the quality of the prior. Structural metrics suggest that the heavy-rain gain reflects recovered heavy rain rather than inflated area, and a matched-ensemble comparison with a reproduced generative baseline indicates that the margin persists.

6 Conclusion

We addressed radar precipitation nowcasting in the high-intensity regime, where existing models blur the strongest convective cells. PG-FMM decouples predictable advection from uncertain small-scale detail and conditions a flow-map head on a frozen physical prior rather than summing a residual onto it, so no conditional-mean blur is additively inherited. Under a unified protocol over four radar benchmarks, PG-FMM surpassed the state-of-the-art AlphaPre on 18 of 24 metrics, with the largest margins at the heavy-rain thresholds where mean-seeking methods fail, and controlled ablations isolated the contributions of the prior and the head.

Two limitations remain. The raw ensemble is under-dispersive, with a spread-to-RMSE ratio of about 0.43 and a U-shaped rank histogram (supplementary material). The probability-matched mean compensates for this only at the verification level, so probabilistic use beyond the reported CRPS and Brier scores requires explicit spread calibration. All results also come from single-seed training on 128×128 rasters under one protocol with lead times up to 100 minutes, and, as with any advection-based nowcaster, PG-FMM under-forecasts initiation and re-intensification without precursors, can under-carry heavy bands (Sec. 5) or over-persist dissipating storms (supplementary material). Addressing these limitations and extending the decoupling to other spatiotemporal tasks are natural next steps.

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Supplementary Material

A Skill versus Lead Time for the Prior Ablation

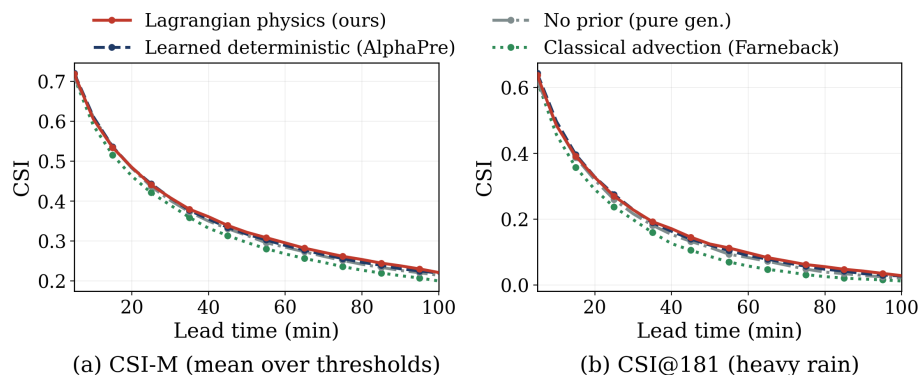


Fig. 6: Skill versus lead time for the prior ablation on SEVIR: CSI averaged over thresholds (a) and at the heavy-rain threshold VIL=181 (b). The margin of the physics prior emerges at long lead times and is largest on heavy rain.

Figure 6 complements the prior ablation (Table 2 of the main paper) with per-lead-time detection skill. All four variants share an identical flow-map head, identical conditioning, and a matched training budget on SEVIR, and are evaluated at the unified operating point (16-member probability-matched mean, four sampling steps); only the content of the prior rollout differs.

Two observations mirror the aggregate table. First, the physics prior is indistinguishable from the alternatives at short lead times, where advection is easy, and its margin appears and grows at longer lead times, the regime where an informative motion prior matters. Second, the classical advection prior degrades fastest, falling below the no-prior variant, which indicates that a poor prior actively misleads the generative head and that the gains of our prior come from its content rather than from conditioning alone.

B Spread–Skill Diagnostic and Sampling-Seed Variability

Figure 7 reports the spread–skill relation of the full model on the SEVIR test set, namely the ensemble spread (standard deviation across the 16 members) and the ensemble-mean RMSE per lead time, in normalized VIL units. The spread grows monotonically with lead time, as the ensemble widens when the forecast becomes less certain, and the spread-to-RMSE ratio is stable at ≈ 0.43 . The ensemble is therefore under-dispersive, the common regime for radar nowcast ensembles, which trade calibration of the raw members for sharpness; this is consistent

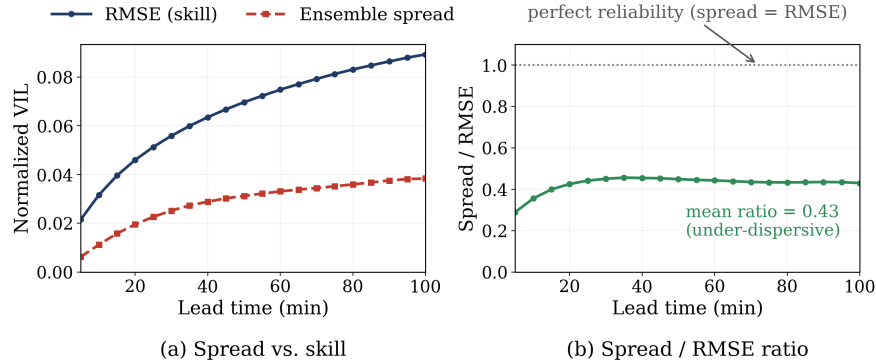


Fig. 7: Spread–skill diagnostic of PG-FMM on SEVIR: ensemble spread and ensemble-mean RMSE per lead time (normalized VIL units). The spread/RMSE ratio is stable at ≈ 0.43 , indicating a sharp but under-dispersive ensemble.

with the strong CRPS of the probability-matched mean. Improving member-level dispersion, for example through spread calibration or larger ensembles, is left as future work.

Sampling-seed variability at the reported operating point, summarized in the main paper, is negligible. Across three independent sampling seeds, the CSI-M standard deviation is 6×10^{-5} and the heavy-rain CSIs vary by 2×10^{-4} (the 16-member probability-matched mean averages out per-sample noise), one to two orders of magnitude below every margin discussed in the main paper.

C Ensemble-Size and Sampling-Step Sweeps

Table 3: Effect of ensemble size on SEVIR (four sampling steps, probability-matched mean). Gains per doubling diminish sharply; 16 members, our default, is the accuracy–compute trade-off. In each column, the best is **bold**, the second best underlined.

Ensemble size	CSI-M $\uparrow$	CSI-181 $\uparrow$	HSS $\uparrow$	MSE $\downarrow$
1	0.3394	0.1667	0.4335	366.80
4	0.3559	0.1816	0.4529	327.32
8	0.3595	0.1843	0.4570	319.65
16 (default)	0.3614	0.1859	0.4593	315.67
32	<u>0.3623</u>	<u>0.1865</u>	<u>0.4603</u>	<u>313.64</u>
64	0.3627	0.1867	0.4607	312.66

Table 3 complements the ablation study of the main paper with a sweep over the ensemble size used by the probability-matched mean, on the same SEVIR

Table 4: Effect of the number of sampling steps N on SEVIR (16-member probability-matched mean fixed). Detection skill saturates: $N=4$ retains 99.8% of the CSI-M of $N=8$ at half the cost per forecast, and even $N=1$ exceeds the strongest deterministic baseline (CSI-M 0.3259). In each column, the best is **bold**, the second best underlined.

Sampling steps	CSI-M $\uparrow$	CSI-181 $\uparrow$	HSS $\uparrow$	MSE $\downarrow$
1	0.3463	0.1590	0.4361	304.50
2	0.3523	0.1698	0.4453	<u>307.27</u>
4 (default)	<u>0.3614</u>	<u>0.1859</u>	<u>0.4593</u>	315.67
8	0.3620	0.1898	0.4616	324.98

checkpoint and at four sampling steps. Every metric improves monotonically with the number of members, but the returns diminish sharply. The CSI-M gain per doubling roughly halves at each step, and moving from 16 to 64 members costs $4\times$ the inference compute for $+0.0013$ CSI-M. We therefore use 16 members throughout the main paper as the accuracy/compute trade-off; few-step sampling makes an ensemble of this size affordable at inference.

Table 4 adds the corresponding sweep over the number of sampling steps N at a fixed 16-member probability-matched mean. One step, the setting most prone to conditional averaging, is the weakest on detection, and multi-step sampling recovers monotonically; $N=4$ is the accuracy-compute operating point used throughout.

D Learned Velocity and Source Fields

E Probabilistic Diagnostics

A representative failure case, selected systematically rather than by hand, is discussed in the main paper together with the qualitative results. Section D below adds further success and failure cases.

F Additional Qualitative Results

Figures 10 and 11 extend the qualitative comparison of the main paper. The SEVIR cases are drawn from the same systematic per-sample CSI-160 ranking as the failure case in the main paper, two successes and one failure, so the gallery is not hand-picked. The additional failure is complementary to the one in the main paper. Instead of under-forecasting a persistent heavy band, the sharp generative head over-persists a dissipating storm that the baseline correctly fades out.

On MeteoNet, no AlphaPre checkpoint is available (its Table 1 entries are transcribed from the AlphaPre paper), so the deterministic reference row is our frozen Lagrangian advection prior, and the panels use the dataset’s discrete reflectivity palette, whose boundaries include the evaluation thresholds

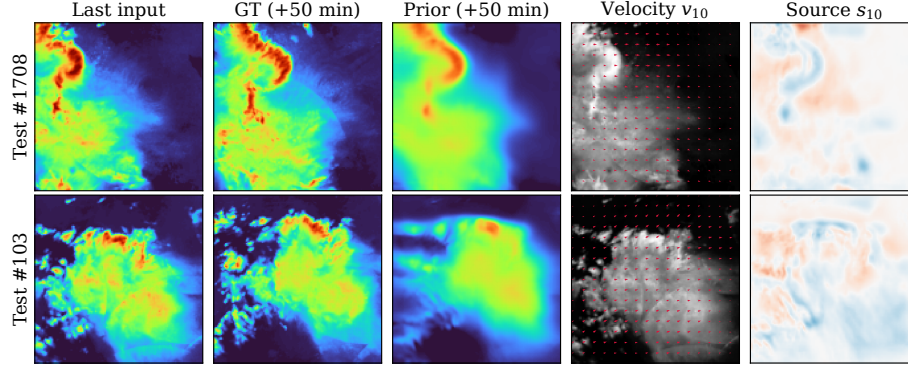


Fig. 8: Learned fields of the frozen Lagrangian prior for two test cases (rows). Columns: last observed frame, ground truth and prior rollout at +50 minutes, the learned velocity v_{10} (arrows, over the last frame), and the learned source s_{10} (diverging colors, ± 0.05 of the normalized intensity). The velocity forms a coherent advective flow, and the source is small and sparse (mean $|s|$ is 1.3% of its bound $c_s=0.25$; 5.6% of pixels exceed 0.01), consistent with its role as a growth/decay correction. Over rain pixels the first-step velocity agrees with Farnebäck optical flow at a cosine similarity of 0.865.

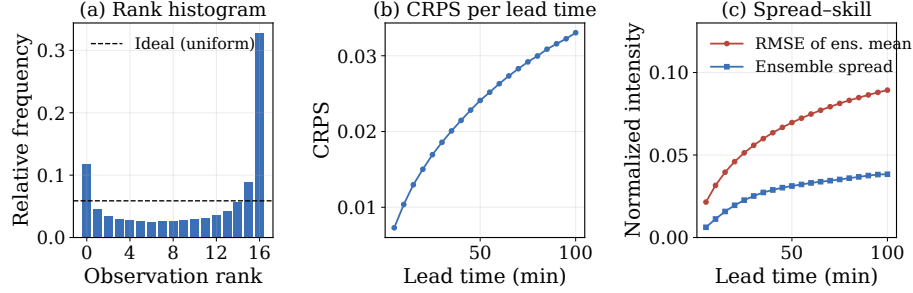


Fig. 9: Probabilistic diagnostics of the 16-member ensemble on the full SEVIR test set. (a) The rank histogram over rain pixels is U-shaped and right-heavy, i.e., the raw ensemble is under-dispersive, consistent with the limitation stated in the conclusion; the probability-matched mean compensates at the verification level. (b) CRPS grows monotonically with lead time. (c) Ensemble spread stays below the RMSE of the ensemble mean (spread/RMSE ≈ 0.43). Brier scores at VIL=16–219 are 0.0583, 0.0361, 0.0151, 0.0058, 0.0027, 0.0005.

$\{12, 18, 24, 32\}$ dBZ. The division of labor matches SEVIR. The prior keeps the system on its observed track but, being a smooth semi-Lagrangian rollout, dilutes the embedded 32–40 dBZ cells into the surrounding band as lead time grows; the generative head restores those heavy cores where the ground truth keeps them (per-case CSI-32 0.338 vs. 0.230 for the prior on index 1138, and 0.309 vs. 0.296 on index 962). In the failure, the observed scattered convection keeps regenerating heavy cells across the domain; the prior never carries them, and the anchored

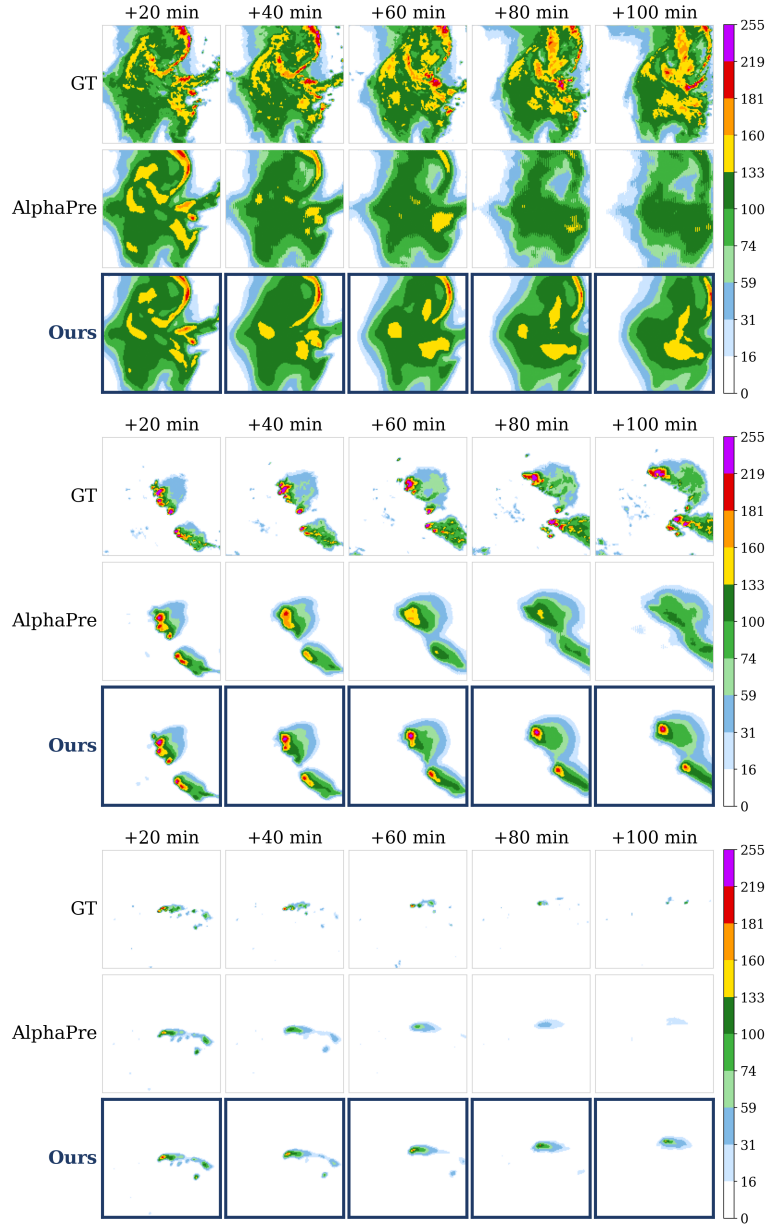


Fig. 10: Additional SEVIR-VIL cases (rows: ground truth, AlphaPre, PG-FMM; columns: +20 to +100 minutes). Top: a success on a frame-filling system (test index 1955; CSI-160 0.178 vs. 0.147). Middle: a success (index 7243; 0.258 vs. 0.221). Bottom: a failure (index 635; 0.196 vs. 0.211), in which the observed storm dissipates by +80 minutes and AlphaPre correctly fades, whereas the generative head maintains a compact storm.

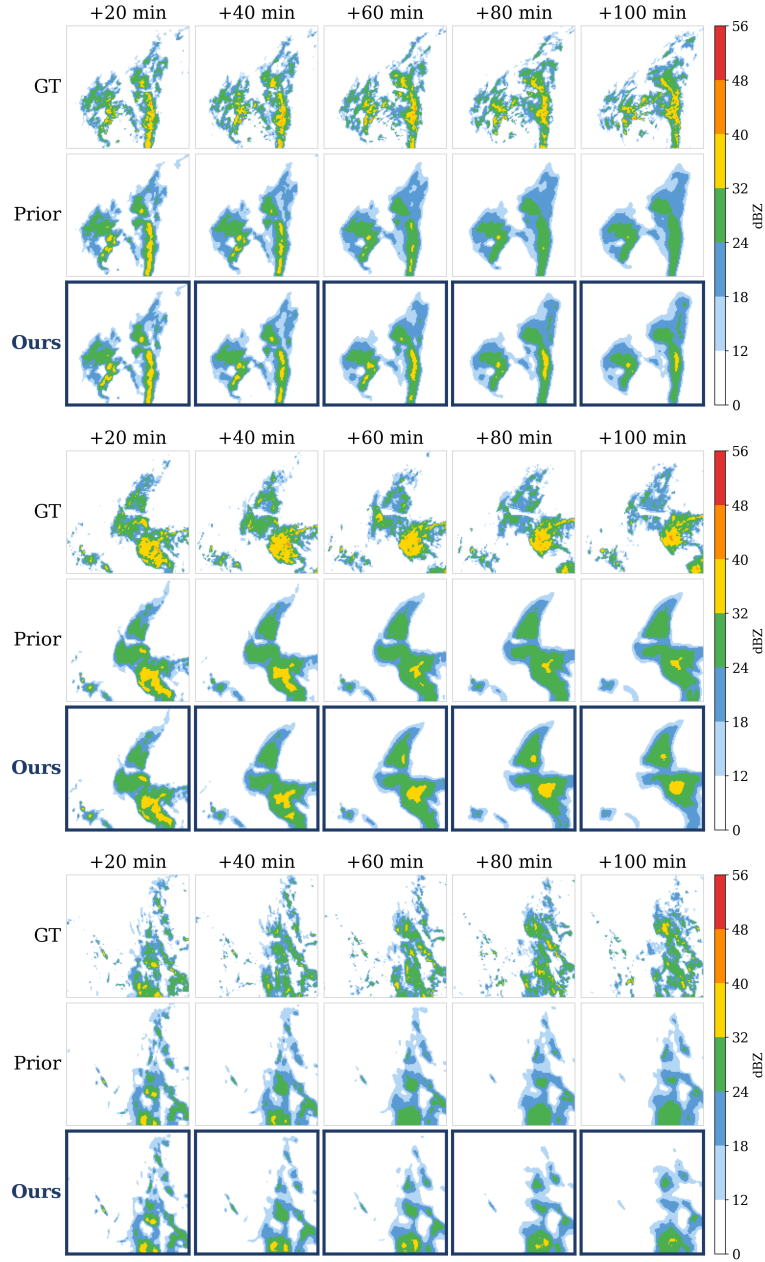


Fig. 11: MeteoNet cases (rows: ground truth, the frozen advection prior, PG-FMM; columns: +20 to +100 minutes; reflectivity in dBZ). Top and middle: successes (test indices 1138 and 962), where the prior smooths the embedded 32–40 dBZ cells away with lead time and PG-FMM restores them. Bottom: a failure (index 460), where the observed convection keeps regenerating heavy cells that the prior never carries and the head only partially restores.

head restores only the strongest core, under-forecasting the rest. Initiation and re-intensification that the advection prior cannot represent remain the hardest regime, a limitation inherited from advection-based nowcasting.

G Implementation Details

Table 5 lists the full set of hyperparameters summarized in Sec. 5.2 of the main paper, grouped by module (optimization, Lagrangian prior, flow-map head, inference, hardware); symbols are defined in Sec. 4. The soft-CSI term of the prior objective (Eq. 6 of the main paper) relaxes, for each evaluation threshold κ of the dataset, the hard exceedance mask to a sigmoid $\sigma((\hat{R} - \kappa)/\alpha)$ with sharpness α , computed on the dataset’s native intensity scale; soft hit, miss, and false-alarm counts are accumulated, and $1 - \text{CSI}_\kappa$ is averaged over the dataset’s thresholds. The warp of Eq. 5 samples on a normalized grid with border padding.

H Composition-Consistency Ablation

Table 6 isolates the effect of the composition-consistency term $\mathcal{L}_{\text{CC}}$ (Eq. 8 of the main paper) on the full model, at its full training schedule. Removing $\mathcal{L}_{\text{CC}}$ slightly raises the detection metrics (CSI-M, CSI-181, CSI-219, HSS) but degrades every fidelity and calibration metric (SSIM, MSE, CRPS). We keep the term enabled for the sharper, better-calibrated forecasts it produces and do not claim it as a source of detection skill; the detection gap is small relative to the margins over prior work reported in the main paper.

I Structural Verification of the Heavy-Rain Gain

CSI is computed after thresholding and is therefore insensitive to boundary detail, so a broad forecast could in principle raise it by covering more area. Table 7 tests that alternative explanation directly with the frequency bias, the ratio of the forecast above-threshold area to the observed one, together with POD, FAR, the relative area error, and multi-scale FSS. Area inflation would require a bias above one. Ours is 0.35 and 0.21 at VIL=181 and 219, so the forecast still covers less heavy-rain area than the observations, and AlphaPre covers even less (0.23 and 0.10). The gain therefore appears to be the recovery of heavy rain that blurred forecasts erase rather than over-prediction, and it is accompanied by the highest POD and the highest FSS at every neighbourhood scale we evaluated (1–33 px). The higher FAR of the single sample is the expected cost of sharp stochastic detail, and CSI already accounts for it.

The table also separates the three operating points discussed in Sec. 4.4 of the main paper. A single sample uses neither the ensemble nor the probability-matched mean and already exceeds AlphaPre. The direct mean of 16 samples falls back to the level of AlphaPre at the heavy thresholds, because averaging cancels displaced peaks, which is the failure mode the probability-matched mean avoids.

Table 5: Implementation details and hyperparameters, grouped by module. Values are given as SEVIR / MeteoNet where they differ by dataset; symbols are defined in Sec. 4 of the main paper.

Hyperparameter	Value
<i>Optimization (both modules)</i>	
Optimizer	AdamW
Learning rate	10^{-4}
Precision	bf16
Gradient clipping	0.5 (head) / 1.0 (prior)
EMA decay ρ	0.999
<i>Lagrangian prior</i>	
Max. displacement $d_{\max}$	8 px
Source scale c_s	0.25
Loss weights $(\lambda_{\text{mse}}, \lambda_{\text{csi}}, \lambda_v, \lambda_s)$	(0.25, 0.02, 0.01, 10^{-3})
Soft-CSI sharpness α	8 (native intensity units)
Batch size / warm-up	16 / 2k steps
Training steps	50k / 30k
<i>Flow-map head</i>	
Base width / channel multipliers	128 / [1, 2, 3, 4]
Residual blocks per level	2
Coarsest resolution	16×16 (self-attention)
Parameters	$\sim 85\text{M}$ ($\sim 101\text{M}$ with frozen prior)
Min. time gap $\delta_{\min}$	0.05
Direct-map probability p_0	0.5
Composition-consistency weight λ_{CC}	$0 \rightarrow 0.04$, steps 12k–30k
Batch size	8
Warm-up	5k steps
Training steps	up to 297.6k / 154k
<i>Inference</i>	
Sampling steps N	4
Ensemble size K	16
<i>Hardware</i>	
GPU	1 $\times$ NVIDIA GB202 (96 GB)
Instance	AWS g7e.4xlarge
Training time (MeteoNet, 154k steps)	~ 11.5 h
Test evaluation (MeteoNet, 16-member)	~ 7 min / 1,310 sequences

J Fusion Ablation: Conditioning versus Additive Residual

Table 8 completes the controlled grid behind the decoupling design by crossing the prior content with the mechanism by which the prior enters the model. With our Lagrangian physics prior the two fusion mechanisms perform comparably, so we do not claim a skill advantage for conditioning in isolation. With

Table 6: Composition-consistency ablation on SEVIR (full model, full training schedule; 16-member PMM, four sampling steps). Removing $\mathcal{L}_{CC}$ slightly raises the detection metrics while degrading every fidelity metric. In each column, the best result is in **bold**.

Variant	CSI-M $\uparrow$	CSI-181 $\uparrow$	CSI-219 $\uparrow$	HSS $\uparrow$	SSIM $\uparrow$	MSE $\downarrow$	CRPS $\downarrow$
without $\mathcal{L}_{CC}$	0.3652	0.1922	0.0973	0.4646	0.7286	318.71	0.0296
with $\mathcal{L}_{CC}$	0.3614	0.1859	0.0925	0.4593	0.7291	315.67	0.0294

Table 7: Structural verification on the full SEVIR test set (8,096 sequences) at the heavy-rain thresholds VIL=181 / 219. FSS uses a 17px neighbourhood, and the area error is the relative error of the above-threshold area. Scores accumulate contingency counts over the test set rather than averaging per-sample CSI as in Table 1 of the main paper, so the values are comparable across rows here but not with that table. In each column, the best result is in **bold**.

Method	CSI $\uparrow$	Bias $\rightarrow$ 1	POD $\uparrow$	FAR $\downarrow$	FSS $\uparrow$	Area err. $\downarrow$
AlphaPre	.157 / .071	.234 / .104	.168 / .073	.284 / .295	.517 / .258	.768 / .899
PG-FMM (single sample)	.170 / .084	.345 / .201	.196 / .093	.434 / .534	.622 / .378	.660 / .812
PG-FMM (mean of 16)	.161 / .071	.240 / .101	.172 / .073	.285 / .283	.537 / .258	.761 / .900
PG-FMM (PMM, default)	.189 / .097	.349 / .213	.214 / .107	.387 / .498	.645 / .411	.654 / .797

Table 8: Fusion ablation on SEVIR. All four variants share the identical flow-map head and conditioning channels at a matched 150k-step budget and are evaluated at the same operating point (16-member PMM, four sampling steps); only the prior content (rows) and the way the prior enters the model (columns) change. Cells give CSI-219 $\uparrow$ / CRPS $\downarrow$.

Prior \ Fusion	Conditioning (ours)	Additive residual
Lagrangian physics (ours)	.0907 / .0298	.0931 / .0297
Deterministic (AlphaPre)	.0892 / .0299	.0846 / .0306

the blurred deterministic prior, however, only additive residual fusion degrades, in both heavy-rain skill and calibration. This is consistent with the residual-anchor failure mode discussed in Sec. 2 of the main paper, reproduced here under a matched budget: summing a conditional-mean forecast onto the output re-imposes its blur, whereas conditioning leaves the head free to replace it. These results indicate that conditioning is the mechanism robust to the quality of the prior, while the prior content remains the factor that decides skill, consistent with the prior ablation in the main paper.

K Matched-Ensemble Comparison with a Generative Baseline

The main comparison reports PG-FMM at its 16-member operating point against baselines that are single deterministic forecasts, so we additionally compare

Table 9: Matched-ensemble comparison on SEVIR. Both methods use eight members aggregated by the same probability-matched mean and are scored by the same evaluator on a fixed subset of the first 512 test sequences, chosen without selection. NFE is the number of network evaluations per member.

Method	NFE / member	CSI-M $\uparrow$	CRPS $\downarrow$
FlowCast (reproduced)	10	.348	.038
PG-FMM	4	.378	.031

against a generative baseline under a matched ensemble and matched post-processing. FlowCast (Sec. 2 of the main paper) is absent from Tab. 1 because its released code and checkpoint cover SEVIR only, at a resolution and horizon that differ from our protocol, so we reproduced it and adapted it to this protocol. Generating a full 16-member ensemble for it takes roughly 72 hours, so both methods are evaluated with eight members on a fixed, unselected subset of the first 512 test sequences. In Table 9, PG-FMM remains ahead in both detection and calibration while using 2.5 times fewer network evaluations per member.